

NetEngine A816 E

Industry's most cost-effective IPv6+ access router

Overview

HUAWEI NetEngine A816 E is an integrated access router oriented to the 5G and cloud era. It is mainly used in government commissions and bureaus, MAN access in villages and towns, financial outlets, railway stations, and enterprise branches. The details are as follows:

- The NetEngine A816 E is 1 U high and 220 mm deep. It provides eight service ports, such as GE and FE ports, to meet the requirements for traffic growth and diversified service access in the cloud era. Network slicing is supported. Multiple types of services can be carried on one network, simplifying network layers.
- NetEngine A816 E supports SRv6 smart connections. SRv6 provides massive consecutive addresses and rich scalability based on IPv6. SRv6 enables automatic cross-domain connections and minute-level service provisioning, implementing cloud-based network scheduling and one-hop access to the cloud. SRv6 can identify applications and tenants and implement intelligent scheduling such as latency and bandwidth based on customer requirements to ensure service SLAs. The number of network protocols is reduced from 10+ to 2, simplifying network O&M.
- The NetEngine A816 E supports intelligent O M by introducing automation and intelligent technologies. Based on the iMaster NCE and iFi technologies, the NetEngine A816 E accurately detects network SLAs in real time, provides real-time service quality visualization, and supports minute-level fault locating.

The NetEngine A816 E integrates multiple functions, simplifies the network structure, provides various service types, reliable service quality, and intelligent O&M. The NetEngine A816 E leads the IP WAN to an intelligent network with autonomous driving, and continuously drives the business success of enterprise customers.

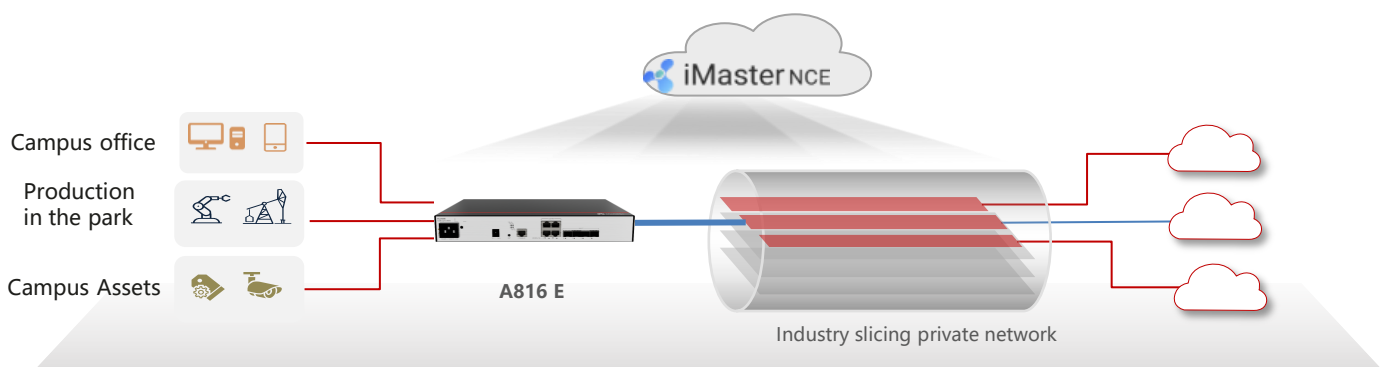


NetEngine A816 E (AC)



NetEngine A816 E (dual ACs)

Application Scenario



Product Highlights

Compact and large capacity

- Size: 300 mm deep cabinet, 1 U high
- Capacity: 120 Gbit/s

IPv6+ fully supported

- SRv6 Ready, Simplified Network
- Network slicing, one network for multiple purposes
- iFi, with-flow detection

Flexible deployment

- Supports mobile phone deployment and remote deployment.
- 1 U high, can be placed on a desktop, wall, or cabinet.
- Flexible switching between MPLS and SRv6 modes

Product Specifications

Property	Description
Interface Type	4*GE/FE (optical) + 4*GE/FE (electrical)
Main control board	Securing the box-shaped structure
Service Slot	Securing the box-shaped structure
Power supply	Single AC, dual AC, single AC+single DC
Dimensions (W x D x H)	320 mm x 220 mm x 43.6 mm (1U)
Typical power consumption	24 W
Heat Dissipation Mode	Natural heat dissipation
Long-term operating temperature	-5°C - 50°C
Voltage Range	AC: 100 V-240 V DC: ±48 V/+24 V to 12 V power
MPLS (MPLS Mode)	LDP, EVPN VLL/VPLS
Segment Routing (SRv6 Mode)	SRv6 BE, SRv6 Policy
Network Slicing	channelized sub-interface
IPsec	300M@512Byte
NAT	600Mbps@512Byte

Product Features

Features	Description
Ethernet Layer 2 Protocol Ethernet service	G.8032, STP/RSTP, MSTP, Layer 2 protocol transparent transmission, BPDU, and LACP VLAN, Bridge Domain, QinQ
Route selection	RIP, RIPng, OSPF, OSPFv3, IS-IS, IS-ISv6, BGP, BGPv6, static routing, static multicast, and BIERv6
IPv4 protocol	ICMP, IP, TCP, UDP, Socket (TCP/UDP/RAW IP), ARP, Static DNS, DNS Client, DHCP Relay, DHCP, Ping, Tracert, NQA

Features	Description
IPv4 protocol	IP policy-based routing Flow-based forwarding next hop IP policy-based routing load balancing IPv4 load balancing is supported.
IPv6 Protocol	IPv6(ND), Path MTU (PMTU) TCP6, Ping IPv6 Tracert IPv6, Socket IPv6 DHCPv6 Relay, Static IPv6 DNS Policy-based routing
MPLS	LDP, MPLS TE, SR MPLS, MPLS-TP OAM
L2VPN	PWE3/VLL, VPLS, EVPN
L3VPN	CE routers can access L3VPN through L3 interfaces. Supports static routes between CEs and PEs, between BGP, RIP, OSPF, and IS-IS. Inter-AS VPN, including RFC 2547bis option A, RFC 2547bis option B, and RFC 2547bis option C
SRv6	SRv6 BE, SRv6 TE Policy, Flex-Algo, G-SRv6, SRv6 network slicing, and anti-microloop
Value-added service	NAT/CGN, IPsec, Xsec
O&M	IFIT, NQA, TWAMP, BFD, Telemetry VRRP, Ethernet OAM, and bit-error-triggered protection switching
QoS	QPPB, DiffServ, level-3 HQoS, redirection, ACL, and policy-based routing
Other	AAA, RADIUS, SSH, SNMP v3, and system logs

Compliance

Project	Description
Compliance	EMC
	• ANSI C63.4
	• AS/NZSPR 32
	• CISPR 24
	• CISPR 32
	• EN 55024
	• EN 55032
	• ETSI EN 300 386
	• FCC CFR47 Part 15 Subpart B
	• ICES-003 Issue 6
	• ICES-GEN Issue 1
	• IEC 61000-3-2
	• IEC 61000-3-3
	• IEC 61000-4-11
	• IEC 61000-4-2
	• IEC 61000-4-29
	• IEC 61000-4-3
	• IEC 61000-4-4
	• IEC 61000-4-5
	• IEC 61000-4-6
	• IEC 61000-6-2
	• IEC 61000-6-4
	• VCCI-CISPR 32
	Environment
	• ETSI EN 300 019-1-1
	• ETSI EN 300 019-1-2
	• ETSI EN 300 019-1-3
	• ETSI EN 300 753
	• ETSI EN 300 019-2-1
	• ETSI EN 300 019-2-2

Project	Description
	• ETSI EN 300 019-2-3 • IEC 60068-2-1 • IEC 60068-2-2 • IEC 60068-2-14 • IEC 60068-2-27 • IEC 60068-2-30 • IEC 60068-2-31 • GB/T 2423.26-2008 • IEC 60068-2-39 • IEC 60068-2-64 • ISTA 2A • ISO 7779 Security. • IEC/EN/UL/CSA 60950-1 • IEC/EN 62368-1 Environmental protection • 2011/65/EU & (EU)2015/863 (EU RoHS) • Regulation (EC) No.1907/2006 (REACH) • 2012/19/EU (WEEE) • 2006/66/EC and 2013/56/EU on batteries and accumulators

NOTE

- With regard to the physical dimensions provided in the table, the width (W) does not consider the mounting ears.
- Temperature and humidity are measured 1.5 meters above the ground and 0.4 meters in front of the cabinet. There is no protection plate on the front or rear side of the cabinet.
- "Short-term" means that the continuous working time does not exceed 96 hours and the accumulated working time does not exceed 15 days per year. If the working time exceeds any of these values, it is considered "long-term".

Software upgrade path

Access to [Software upgrade path](#) Obtain the VRP software version or patch.

More information

For more information about NetEngine A800 E-Series Routers, visit <https://e.huawei.com> or contact us by:

- Global service hotline : <https://e.huawei.com/cn/service-hotline>
- Log in to Huawei enterprise technical support website : <https://support.huawei.com/enterprise/>
- Customer service email : support_e@huawei.com

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